

VAASA KLEMETTILA, Finland

WMO: 029570

Lat: 63.0989N

Lon: 21.6394E

Elev: 9

StdP: 101.22

Time Zone: 2.00 (EUE)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.9	-17.7	-23.4	0.5	-20.8	-20.1	0.6	-17.5	11.8	1.3	10.7	0.6	2.3	120	0.660

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.0	25.4	17.9	23.4	17.1	21.7	16.2	19.4	23.1	18.1	22.0	17.1	20.7	4.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.9	12.9	21.5	16.6	11.9	19.9	15.5	11.0	18.7	55.2	23.0	51.3	22.2	48.0	20.6	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.8	7.8	-21.8	27.4	6.1	2.0	-26.2	28.9	-29.7	30.1	-33.2	31.2	-37.6	32.7		
(5)			-22.1	20.2	5.8	1.7	-26.3	21.4	-29.7	22.3	-32.9	23.3	-37.2	24.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.1	-5.5	-5.2	-2.5	3.1	9.5	13.6	17.1	15.4	11.3	5.3	1.0	-2.5	(6)
(7)		DBStd	8.86	5.96	6.25	4.58	3.39	4.15	2.98	3.01	2.62	2.98	3.36	4.12	5.10	(7)
(8)		HDD10.0	2389	479	425	388	208	59	4	0	1	21	147	270	387	(8)
(9)		HDD18.3	4861	737	658	647	456	276	146	62	99	212	403	520	645	(9)
(10)		CDD10.0	603	0	0	0	2	43	111	219	167	59	3	0	0	(10)
(11)		CDD18.3	34	0	0	0	0	2	3	23	7	0	0	0	0	(11)
(12)		CDH23.3	187	0	0	0	0	13	13	130	29	1	0	0	0	(12)
(13)		CDH26.7	28	0	0	0	0	1	0	24	3	0	0	0	0	(13)
(14)	Wind	WSAvg	3.9	3.9	4.1	4.0	3.8	3.9	4.0	3.6	3.7	3.5	3.8	3.9	4.1	(14)
(15)	Precipitation	PrecAvg	506	32	23	25	24	34	42	60	66	58	53	49	40	(15)
(16)		PrecMax	708	70	60	55	66	77	119	152	180	108	102	114	96	(16)
(17)		PrecMin	335	6	2	6	0	3	3	5	5	16	17	9	7	(17)
(18)		PrecStd	87	17	14	13	16	19	22	40	38	27	23	22	22	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.6	5.6	8.0	17.1	25.1	25.0	29.4	26.6	20.5	14.1	10.2	6.7	(19)
(20)			MCWB	3.4	3.4	4.5	9.3	15.5	16.6	19.2	19.8	15.8	11.4	8.4	5.8	(20)
(21)		2%	DB	3.5	4.3	5.8	12.2	22.0	22.7	26.6	23.6	18.3	12.3	8.2	5.6	(21)
(22)			MCWB	2.4	2.7	3.2	7.5	13.9	16.1	18.6	17.7	14.5	10.9	7.0	4.8	(22)
(23)		5%	DB	2.5	3.0	4.5	10.2	19.2	20.7	24.5	21.6	16.9	11.1	7.0	4.6	(23)
(24)			MCWB	1.6	1.9	2.4	6.3	12.6	14.7	17.9	16.4	13.6	9.8	6.2	3.8	(24)
(25)		10%	DB	1.5	2.0	3.3	8.5	16.2	19.0	22.5	19.9	15.7	10.0	6.1	3.4	(25)
(26)			MCWB	0.9	1.2	1.7	5.2	10.9	14.0	17.2	15.8	13.1	8.8	5.5	2.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.9	3.9	4.9	10.5	16.8	18.5	21.1	20.9	16.6	12.4	8.8	6.3	(27)
(28)			MCDWB	4.5	5.4	7.6	14.9	23.0	22.9	26.0	25.2	19.2	13.4	10.3	6.7	(28)
(29)		2%	WB	2.6	2.7	3.6	8.0	14.9	16.9	19.8	18.7	15.4	11.1	7.2	5.0	(29)
(30)			MCDWB	3.3	3.8	5.2	11.8	20.4	20.5	23.8	21.8	17.4	12.0	7.9	5.5	(30)
(31)		5%	WB	1.7	2.1	2.8	6.6	12.9	15.7	18.9	17.4	14.4	10.1	6.3	3.7	(31)
(32)			MCDWB	2.3	2.9	4.2	9.7	18.4	19.4	22.7	20.6	16.0	10.9	6.8	4.3	(32)
(33)		10%	WB	1.0	1.3	1.8	5.4	11.2	14.6	17.8	16.3	13.6	9.0	5.5	2.7	(33)
(34)			MCDWB	1.5	1.9	3.2	8.0	15.4	18.1	21.6	18.9	15.2	9.9	6.1	3.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.9	5.3	6.5	7.0	7.6	6.9	7.0	6.7	6.1	4.5	3.3	4.2	(35)
(36)			MCDBR	3.9	4.9	6.0	10.5	12.2	10.2	10.2	9.7	7.9	5.1	3.7	3.9	(36)
(37)			MCWBR	3.5	4.2	4.1	6.2	5.8	5.1	4.4	4.7	4.8	4.1	3.5	3.5	(37)
(38)		5% WB	MCDWB	3.8	4.7	5.3	9.1	11.2	8.7	8.7	7.9	6.6	4.6	3.5	3.8	(38)
(39)			MCWBR	3.6	4.2	4.0	5.8	5.6	4.8	4.3	4.4	4.4	4.0	3.4	3.6	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub	0.240	0.296	0.319	0.357	0.355	0.353	0.374	0.361	0.335	0.318	0.381	0.350	(41)	
(42)		taud	2.064	2.190	2.276	2.319	2.433	2.469	2.433	2.462	2.516	2.381	2.663	2.554	(42)	
(43)		Ebn at Noon	384	632	777	823	861	870	837	816	769	614	354	211	(43)	
		Edh at Noon	31	63	85	101	98	97	98	88	69	53	29	18		
(44)	All-Sky Solar Radiation	RadAvg	0.15	0.71	2.22	3.82	5.26	5.89	5.39	4.11	2.41	0.92	0.22	0.07	(44)	
(45)		RadStd	0.02	0.14	0.26	0.39	0.50	0.44	0.47	0.36	0.23	0.18	0.04	0.01	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (41 neighbors)	+0.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page